

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION OCTOBER 2025



Course: B. Tech
Subject Name: EMEC-I
Maximum Marks: 20

Semester: III
Subject Code: EE-24212
Date: 07-10-2025
Time: 01 Hour

Instructions: All questions are compulsory. Draw phasor diagrams wherever necessary.

Q. No.	Questions	Marks	COs
1.	On a 415 V system, two watt-meters show a ratio $W_1:W_2=3:1$ (both positive). If the total power is 30 kW, find: (i) power factor, (ii) line current, (iii) W_1 and W_2	05	CO 1
2.	With phasor diagrams, show how balanced three-phase input produces two secondary voltages displaced by 90° in a Scott connection. Also, derive the expressions for: (a) the three-phase primary line currents on the supply side of a Scott-connected transformer bank, and (b) the two-phase secondary currents on the load side, assuming the two-phase load is balanced. Clearly state any assumptions you make about the transformer ratios and connections.	05	CO 2
3.	A 25 kVA, single phase transformer rated 2000/400 V at no load, has $R_1=5$ ohms, $X_1=10$ ohms, $R_2=0.2$ ohms, $X_2=0.5$ ohms. Determine the approximate value of the secondary voltage at full load, 0.8 pf (lagging), when the primary applied voltage is 2000 V.	05	CO2
4.	A three-phase step down transformer is connected to 6600 V on the primary side. The ratio of turns per phase is 10 and the line current drawn from the mains is 25 A. Find secondary line voltage, line current and output if the transformer is (a) star-star (b) delta-delta (c) star-delta and (d) delta-star. Neglect losses.	05	CO2